

VAAHS Procedural Antibiotic Guidelines for Interventional Radiology

The goals of this guideline are to decrease the incidence of post-procedural infection by ensuring optimal antibiotic selection and dosing, and to ensure antimicrobial prophylaxis is only used when medically indicated to prevent the development of resistant organisms and adverse effects to the patient. General antibiotic prophylaxis recommendations are provided in the chart below. Recommendations are based on type of procedure and patient-specific risk factors. Other circumstances, such as history of infections relevant to the area of the procedure may result in a need to expand coverage beyond these recommendations. Additional information regarding antibiotic dosing and allergy evaluations are provided at the end of this document. Please contact ID pharmacist for further recommendations (Ron Kendall, x55829).

PROCEDURE	INFECTION RISK & RISK FACTORS TO CONSIDER	ANTIBIOTIC RECOMMENDATIONS
Diagnostic Angiography and Angioplasty	Low incidence of infection, only give antibiotic prophylaxis if risk factor(s) for infection: long procedure duration, multiple catheterizations at same site, difficult arterial access, post-procedure maintenance of an arterial sheath, local bleeding, or congestive heart failure.	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Bare metal stent placement	Very low incidence of infection, only give antibiotic prophylaxis if risk factor(s) for infection: advanced age, CKD, diabetes, immunosuppression, long procedures with multiple guidewire exchanges, or placement of indwelling catheters in place > 6 hours.	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Arterial Endografts	Very low incidence of infection, only give antibiotic prophylaxis if performed in an emergency setting	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Arteriovenous Fistula and Graft Interventions	Very low incidence of infection, only give antibiotic prophylaxis if placing a covered stent	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Closure Devices	Very low incidence of infection, only give antibiotic prophylaxis if patient is obese, diabetic, and with prior closure device placement in the previous 6 months	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Uterine Artery Embolization	Infectious complications up to 2% of cases, give all patients prophylaxis	Preferred: Cefazolin OR Ampicillin/Sulbactam
Hepatic Embolization and Chemoembolization	Infectious complications are rare in patients with a competent sphincter of Oddi. Patient with incompetent sphincter of Oddi (history of sphincterotomy, biliary stent placement, or bilioenteric anastomosis) are high risk for infection at 5-25% incidence	Preferred: Ampicillin/Sulbactam OR Cefoxitin OR Cefazolin +/- Metronidazole OR Ceftriaxone +/- Metronidazole Patients with incompetent sphincter of Oddi should receive oral antibiotics for 5d post-procedure with Amoxicillin/clavulanate or Cefuroxime + Metronidazole

Radioembolization	Very low incidence of infection, routine prophylaxis not recommended	Preferred: None
Gastrointestinal Embolization	Low incidence of infection, routine prophylaxis not recommended except in cases of hemobilia in which accumulation of blood can lead to cholangitis	Preferred: None Qualifying risk factor: Ceftriaxone OR Ampicillin/Sulbactam
Splenic artery embolization	Risk of infection depends on extent of ischemia created, routine prophylaxis not recommended unless >70% of spleen is expected to be embolized	Preferred: None Qualifying risk factor: Ceftriaxone or Ampicillin/Sulbactam
Central Venous Catheter Placement	Risk of systemic infection is rare, between 0.4 and 3% with multiple trials failing to demonstrate prevention of infection by giving antibiotic prophylaxis. If a patient is receiving CVC for initiation of chemotherapy it is unclear if the added risk of their pending immunocompromise warrants antibiotic prophylaxis.	Preferred: None Qualifying risk factor present: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Tunneled or Non-Tunneled Dialysis Catheter Placement	Risk of systemic infection is rare, but it is unclear if antibiotic prophylaxis is warranted. Consider prophylaxis if patient is immunocompromised.	Preferred: None OR Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Inferior Vena Cava Filter Placement or Retrieval	Very low incidence of infection, routine prophylaxis is not recommended	Preferred: None
Vascular Malformation Treatment	Very low incidence of infection, routine prophylaxis is not recommended unless mucosa of oropharyngeal or gastrointestinal tract is disrupted	Preferred: None OR Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Varicocele embolization	No incidence of infection, prophylaxis is not recommended	Preferred: None
Transjugular Intrahepatic Portosystemic Shunt Creation	Transient bacteremia is common (up to 35%) though infection of the stent lumen is rare. Prophylaxis is recommended.	Preferred: Ceftriaxone OR Ampicillin/Sulbactam
Percutaneous Transhepatic Biliary Drains and Cholecystostomy Tube Placement	Area of drain or tube insertion is frequently infected so prophylaxis is always recommended unless patient is already on broad-spectrum antibiotic therapy for treatment of the intra-abdominal infection.	Preferred: Ceftriaxone OR Ampicillin/Sulbactam (Review any recent abdominal fluid cultures to help guide antibiotic therapy!)
Biliary tube exchange	Usually tube is colonized with bacteria, which may result in bloodstream infection during exchange, so prophylaxis is always recommended.	Preferred: Ceftriaxone OR Ampicillin/Sulbactam
Percutaneous Nephrostomy Tube placement or Exchange, Nephroureteral Stent Placement or exchange	If patient has a urinary tract infection, antibiotic prophylaxis is always recommended unless patient is already on antibiotics for treatment. If patient does not have a urinary tract infection, it is unclear if antibiotic prophylaxis is necessary but consider if patients have advanced age, diabetes, bladder dysfunction, neurogenic bladder, or previous ureteral manipulation. Patients having tube exchange should receive antimicrobial prophylaxis as bacterial colonization of the urinary tract is common.	Preferred: Ceftriaxone (Review any recent urine cultures to help guide antibiotic therapy!)

Gastrostomy or Jejunostomy Tube Placement	Peristomal infection is more common with antegrade technique than retrograde technique but recommended to use antibiotic prophylaxis for both.	Preferred: Cefazolin
Liver Tumor Ablation	Patients without biliary-enteric anastomosis have a low risk for infection but for procedures with a large amount of necrotic material the patient may benefit from prophylaxis against skin flora. There is a higher risk for infection and abscess formation in patients with a history of biliary colonization due to incompetent sphincter of Oddi.	Low Risk: Cefazolin Higher Risk: Ceftriaxone Patients with incompetent sphincter of Oddi may receive oral antibiotics for 5d post-procedure with Augmentin or Cefuroxime + Metronidazole
Renal Tumor Ablation	Infectious complications are rare. Consider antibiotic prophylaxis in high-risk patients with ileal conduit, urinary diversion, or immunocompromise. For procedures with a large amount of necrotic material created the patient may benefit from prophylaxis against skin flora.	Low Risk: Cefazolin High Risk: Ceftriaxone
Percutaneous Abscess Drainage	Patients should already be on systemic antibiotic therapy prior to drainage to prevent rupture and spillage of abscess cavity into surrounding space.	Recommended antibiotic depends on the location and etiology of the abscess, contact ID for recommendations if patient not on antibiotics yet.
Paracentesis and thoracentesis	Clean procedures with infectious complication rates <1%.	Preferred: None
Kyphoplasty	Infectious complications are infrequent but carry high morbidity so antibiotic prophylaxis is recommended.	Preferred: Cefazolin Known MRSA+ or unable to tolerate cefazolin: Vancomycin
Suprapubic Catheter Removal	Antimicrobial prophylaxis at the time of catheter removal has been shown to decrease the prevalence of symptomatic UTIs with a number needed to treat of 17. Consider using prophylaxis, especially in high-risk patients such as renal transplant recipients.	Preferred: Ceftriaxone (Review any recent urine cultures to help guide antibiotic therapy!)

Allergy evaluation:

What was allergic reaction?	Recommended Action
Intolerance or adverse effect to beta-lactam such as headache or GI symptoms	Ok to give patient any beta-lactam antibiotic
Suspected IgE-mediated allergic reaction to beta-lactam including pruritis, mild-to-moderate rash including hives, angioedema, respiratory symptoms, and anaphylaxis. Distant childhood or adolescent reaction to penicillin with no known additional information	Ok to give patient a beta-lactam antibiotic from a different class (i.e. penicillins vs cephalosporins) with a dissimilar molecular structure on its side chain(s). - Cefazolin is not similar in structure to any other beta-lactam. - Ceftriaxone is similar to cefepime, cefotaxime, and cefpodoxime. - Ampicillin is similar to cefaclor and cephalixin.
Severe cutaneous non-IgE mediated allergic reaction such as Stevens-Johnson Syndrome,	Avoid all beta-lactams, contact ID pharmacist for recommendations about alternative antibiotics.

Toxic Epidermal Necrolysis, Drug Rash with Eosinophilia and Systemic Symptoms (DRESS), or Acute Generalized Exanthematous Pustulosis	
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Antibiotic Dosing

Cefazolin	2g if <120 kg, 3g if >=120 kg
Ceftriaxone	2g all patients
Ampicillin/sulbactam	3g all patients
Vancomycin	10-15 mg/kg (actual body weight) max 1.5g

Last Update: 3/16/23